

07 BANK DATABASE – SQL VIEWS

2520030066

Mohammad Rehan

1. Customer + Account View

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: 06-TRIGGERS-STORED-PROC... 07 BANK DATABASE - SQL VIE... x

Limit to 1000 rows

SCHEMAS

Filter objects

- bankdb
 - Tables
 - Views
 - Stored Procedures
 - Functions
- bookflow_db
- sys

Administration Schemas

Information

No object selected

```
52 A.Account_No,  
53 A.Account_Type,  
54 A.Balance  
55 FROM Customer C  
56 JOIN Account A  
57 ON C.Customer_ID = A.Customer_ID;  
58  
59 • SELECT * FROM Customer_Account_View;
```

Result Grid

	Customer_ID	Customer_Name	Account_No	Account_Type	Balance
▶	101	Mohammad Rehan	10001	Savings	55000.00
	102	Akhil Karuturi	10002	Savings	95000.00
	103	Rohit Upputuri	10003	Current	100000.00
	104	Pratik Kumar	10004	Savings	50000.00
	105	Yugandar Peddi	10005	Current	80000.00
	106	Sreekar Varanasi	10006	Savings	77000.00
	107	Arnav Chitkena	10007	Current	175000.00
	108	Keith Dudemeister	10008	Savings	30000.00
	109	Molly Clock	10009	Savings	105000.00
	110	Aditya Kumar	10010	Current	95000.00

Customer_Account_View 3 x

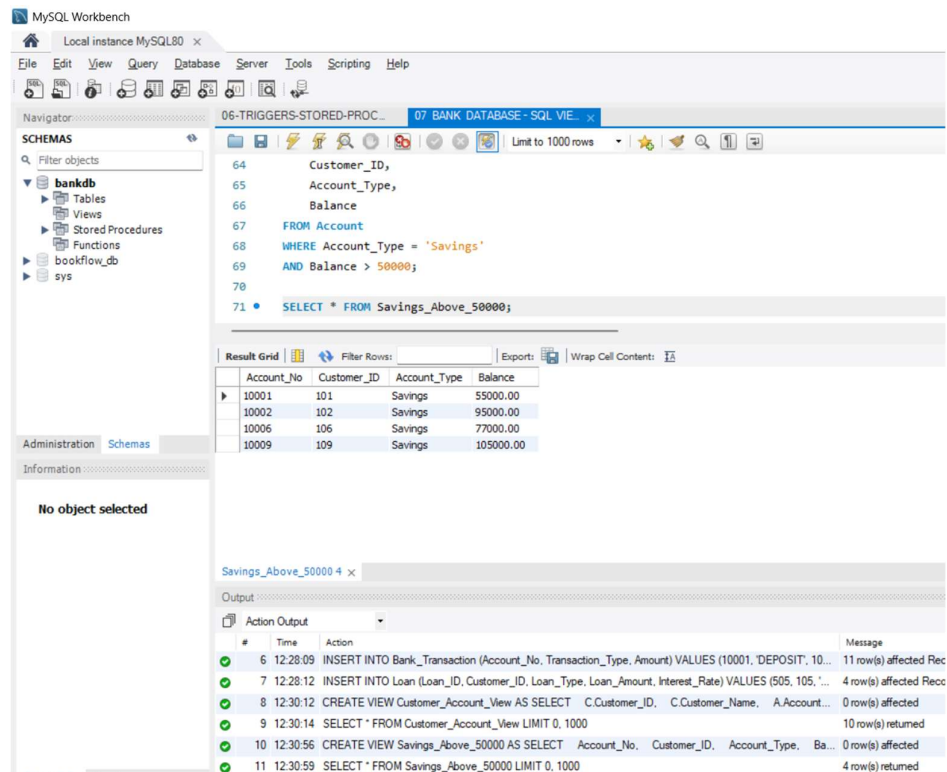
Output

Action Output

#	Time	Action	Message
✓	4 12:28:05	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (106, 'Sreekar Varanasi'...	5 row(s) affected Records: 5 Dupli
✓	5 12:28:07	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10006, 106, '...	5 row(s) affected Records: 5 Dupli
✓	6 12:28:09	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	11 row(s) affected Records: 11 Du
✓	7 12:28:12	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (505, 105, '...	4 row(s) affected Records: 4 Dupli
✓	8 12:30:12	CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, A.Account...	0 row(s) affected
✓	9 12:30:14	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned

Object Info Session

2. Savings Accounts > ₹50,000



MySQL Workbench interface showing a query executed on the 'BANK DATABASE'.

Query:

```
64 Customer_ID,  
65 Account_Type,  
66 Balance  
67 FROM Account  
68 WHERE Account_Type = 'Savings'  
69 AND Balance > 50000;  
70  
71 • SELECT * FROM Savings_Above_50000;
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance
10001	101	Savings	55000.00
10002	102	Savings	95000.00
10006	106	Savings	77000.00
10009	109	Savings	105000.00

Output:

6 12:28:09 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10... 11 row(s) affected Rec

7 12:28:12 INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (505, 105, '... 4 row(s) affected Rec

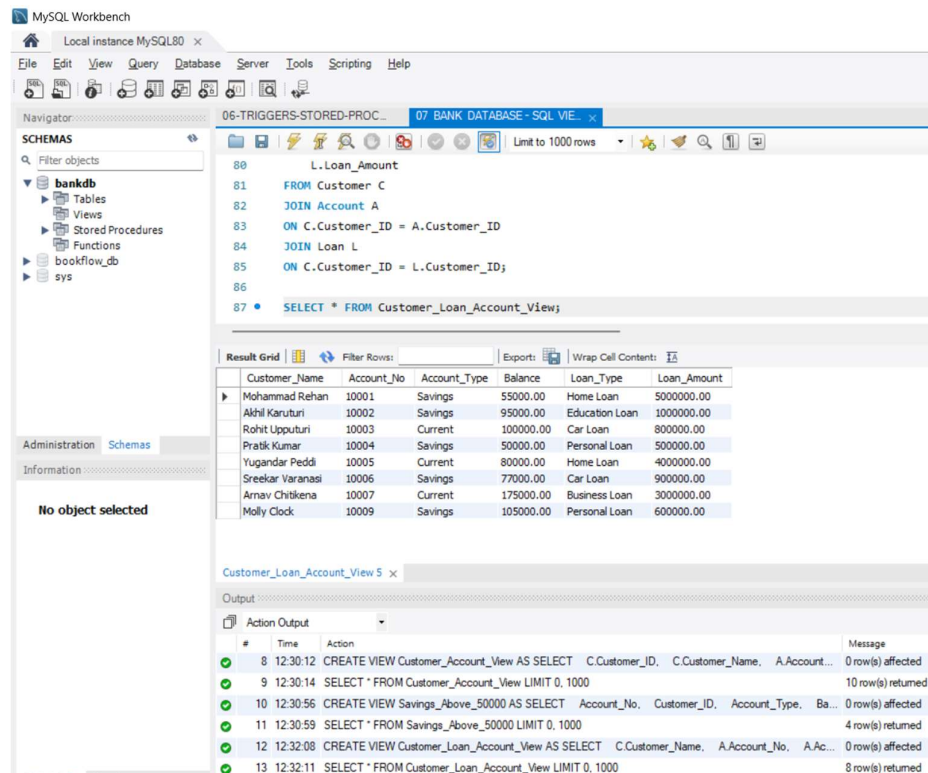
8 12:30:12 CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, A.Account... 0 row(s) affected

9 12:30:14 SELECT * FROM Customer_Account_View LIMIT 0, 1000 10 row(s) returned

10 12:30:56 CREATE VIEW Savings_Above_50000 AS SELECT Account_No, Customer_ID, Account_Type, Ba... 0 row(s) affected

11 12:30:59 SELECT * FROM Savings_Above_50000 LIMIT 0, 1000 4 row(s) returned

3. Customer + Account + Loan View



MySQL Workbench interface showing a query executed on the 'BANK DATABASE'.

Query:

```
80 L.Loan_Amount  
81 FROM Customer C  
82 JOIN Account A  
83 ON C.Customer_ID = A.Customer_ID  
84 JOIN Loan L  
85 ON C.Customer_ID = L.Customer_ID;  
86  
87 • SELECT * FROM Customer_Loan_Account_View;
```

Result Grid:

Customer_Name	Account_No	Account_Type	Balance	Loan_Type	Loan_Amount
Mohammad Rehan	10001	Savings	55000.00	Home Loan	5000000.00
Akhil Karuturi	10002	Savings	95000.00	Education Loan	1000000.00
Rohit Upputuri	10003	Current	100000.00	Car Loan	800000.00
Pratik Kumar	10004	Savings	50000.00	Personal Loan	500000.00
Yugandar Peddi	10005	Current	80000.00	Home Loan	4000000.00
Sreekar Varanasi	10006	Savings	77000.00	Car Loan	900000.00
Arnav Chitkena	10007	Current	175000.00	Business Loan	3000000.00
Molly Clock	10009	Savings	105000.00	Personal Loan	600000.00

Output:

8 12:30:12 CREATE VIEW Customer_Account_View AS SELECT C.Customer_ID, C.Customer_Name, A.Account... 0 row(s) affected

9 12:30:14 SELECT * FROM Customer_Account_View LIMIT 0, 1000 10 row(s) returned

10 12:30:56 CREATE VIEW Savings_Above_50000 AS SELECT Account_No, Customer_ID, Account_Type, Ba... 0 row(s) affected

11 12:30:59 SELECT * FROM Savings_Above_50000 LIMIT 0, 1000 4 row(s) returned

12 12:32:08 CREATE VIEW Customer_Loan_Account_View AS SELECT C.Customer_Name, A.Account_No, A.Ac... 0 row(s) affected

13 12:32:11 SELECT * FROM Customer_Loan_Account_View LIMIT 0, 1000 8 row(s) returned

4. Branch-Wise Total Balance

06-TRIGGERS-STORED-PROC... 07 BANK DATABASE - SQL VIE...

```
89 • CREATE VIEW Branch_Total_Balance_Practical AS
90 SELECT
91     Branch,
92     SUM(Balance) AS Total_Balance
93 FROM Account
94 GROUP BY Branch;
95
96 • SELECT * FROM Branch_Total_Balance_Practical;
```

Result Grid

Branch	Total_Balance
Hyderabad	335000.00
Bangalore	100000.00
Chennai	50000.00
Delhi	77000.00
Mumbai	175000.00
Pune	30000.00
Vijayawada	95000.00

Branch_Total_Balance_Practical 6 x

Output

Action Output

#	Time	Action	Message
✓ 10	12:30:56	CREATE VIEW Savings_Above_50000 AS SELECT Account_No, Customer_ID, Account_Type, Ba...	0 row(s) affected
✓ 11	12:30:59	SELECT * FROM Savings_Above_50000 LIMIT 0, 1000	4 row(s) returned
✓ 12	12:32:08	CREATE VIEW Customer_Loan_Account_View AS SELECT C.Customer_Name, A.Account_No, A.Ac...	0 row(s) affected
✓ 13	12:32:11	SELECT * FROM Customer_Loan_Account_View LIMIT 0, 1000	8 row(s) returned
✓ 14	12:32:44	CREATE VIEW Branch_Total_Balance_Practical AS SELECT Branch, SUM(Balance) AS Total_Balance ...	0 row(s) affected
✓ 15	12:32:46	SELECT * FROM Branch_Total_Balance_Practical LIMIT 0, 1000	7 row(s) returned

5. Annual Loan Interest

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

06-TRIGGERS-STORED-PROC... 07 BANK DATABASE - SQL VIE...

```
101 Customer_ID,
102 Loan_Type,
103 Loan_Amount,
104 Interest_Rate,
105 (Loan_Amount * Interest_Rate / 100) AS Annual_Interest
106 FROM Loan;
107
108 • SELECT * FROM Loan_Interest_Practical;
```

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate	Annual_Interest
501	101	Home Loan	500000.00	7.50	375000.00000000
502	102	Education Loan	100000.00	6.50	65000.00000000
503	103	Car Loan	800000.00	8.20	65600.00000000
504	104	Personal Loan	500000.00	10.50	52500.00000000
505	105	Home Loan	400000.00	7.20	288000.00000000
506	106	Car Loan	900000.00	8.50	76500.00000000
507	107	Business Loan	300000.00	9.00	270000.00000000
508	109	Personal Loan	600000.00	10.00	60000.00000000

Loan_Interest_Practical 7 x

Output

Action Output

#	Time	Action	Message
✓ 12	12:32:08	CREATE VIEW Customer_Loan_Account_View AS SELECT C.Customer_Name, A.Account_No, A.Ac...	0 row(s) affected
✓ 13	12:32:11	SELECT * FROM Customer_Loan_Account_View LIMIT 0, 1000	8 row(s) returned
✓ 14	12:32:44	CREATE VIEW Branch_Total_Balance_Practical AS SELECT Branch, SUM(Balance) AS Total_Balance ...	0 row(s) affected
✓ 15	12:32:46	SELECT * FROM Branch_Total_Balance_Practical LIMIT 0, 1000	7 row(s) returned
✓ 16	12:33:39	CREATE VIEW Loan_Interest_Practical AS SELECT Loan_ID, Customer_ID, Loan_Type, Loan_Am...	0 row(s) affected
✓ 17	12:33:41	SELECT * FROM Loan_Interest_Practical LIMIT 0, 1000	8 row(s) returned